

Jie Liu

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RESEARCH INTERESTS

• **AI Infrastructure** • **Compilers** • **Domain-Specific Languages** • **Heterogeneous Hardware**

My research enables automatic optimization of tensor computations on heterogeneous platforms through tensor-layout abstractions and compiler-based techniques.

WORK EXPERIENCE

Meta Platforms, Inc.

Menlo Park, CA, USA

- Research Scientist (Triton, GPU Kernel Performance)

May 2025 – Present

EDUCATION

Cornell University

Ithaca, NY, USA

- Ph.D. in Computer Systems Lab, ECE

Aug 2019 – May 2025

- Advisor: Prof. Zhiru Zhang

- Courses: Computer Architecture (A), Complex Digital ASIC Design (A), Advanced Compilers (A), Computer Systems Language (Teaching Assistant)

Tsinghua University

Beijing, China

- B.Eng in Microelectronic Science and Engineering

Sep 2015 – Jul 2019

INTERNSHIP EXPERIENCE

Microsoft Research

Redmond, WA, USA

- Research Intern at the Database Group

May 2024 – Aug 2024

Pacific Northwest National Laboratory (PNNL)

Richland, WA, USA

- PhD Intern in the High-Performance Computing (HPC) Group

May 2023 – Aug 2023

AMD Xilinx, Inc.

San Jose, CA

- Compiler Intern at Xilinx Research Labs

May 2021 – Aug 2021

University of California, Los Angeles (UCLA)

Los Angeles, CA, USA

- Undergraduate Research Intern at the VAST Laboratory
- Advisor: Prof. Jason Cong

Jun 2018 – Sep 2018

SKILLS

Frameworks: MLIR / LLVM, Triton, PyTorch

Programming Languages: C/C++, Python, MATLAB, Verilog / PyMTL, $\LaTeX$, SQL, Spice

CAD Tools: Mentor Catapult HLS, Cadence Stratus HLS, Xilinx Vitis HLS

PROJECTS

Automatic Warp Specialization Support in the Triton Compiler

Research Scientist, Triton, Meta

Sep 2025 – Present

- Implemented a ping-pong barrier auto-insertion pass that delivers $\sim 5\%$ speedup on the flash-attention kernel on Blackwell and $\sim 4\%$ on Hopper.
- Identified a performance gap caused by back-to-back MMA ops and resolved it by extending the TLX (Triton Language Extension) API for explicit control over asynchronous MMA execution, yielding $\sim 5\%$ throughput gains on jagged and regular flash-attention kernels.
- Led Triton compiler support for the causal flash-attention kernel, improving throughput from $\sim 28\%$ to $\sim 80\%$ of cuDNN 9.19, and reaching $\sim 89\%$ with additional manual IR rewriting for P/V subtiling.
- Led Triton compiler support for the scaled_mm kernel, reaching parity with the vLLM CUTLASS kernel for TensorWise and RowWise scaling modes, and $\sim 104\%$ of its performance for BlockWise scaling via a hand-written TLX kernel.

Helion: A Python-embedded DSL for Higher-level Triton Kernel Authoring

Research Scientist, PyTorch, Meta

Aug 2025 – Sep 2025

- Authored high-performance Helion kernels and benchmarked them against Tritonbench and torch.compile.

An LLM-based Agent for Triton Kernel Performance Optimization

Research Scientist, Triton, Meta

May 2025 – Sep 2025

- Developed a RAG-enhanced LLM agent to automate Triton kernel rewriting for performance optimization.

- Built a web interface supporting conversational, user-guided iterative Triton kernel optimization.
- Designed a benchmark suite and evaluation framework to assess the agent’s code-generation quality.

AsyncSparse: Accelerating Sparse Matrix-Matrix Multiplication on Asynchronous GPU Architectures

Ph.D. candidate, with Prof. Zhiru Zhang, Cornell University Sep 2024 – May 2025

- Co-designed high-performance SpMM kernels for NVIDIA Hopper’s asynchronous execution model using the Block-CSR and Window-CSR sparse formats.
- Built a warp-specialized producer-consumer pipeline overlapping TMA data movement with WGMMMA computation, plus a load-balanced WCSR kernel that splits large row-windows into fixed-size tasks.
- Achieved geomean speedups of $1.47\times$ over SOTA AccSpMM and $6.24\times$ over cuSPARSE across SuiteSparse matrices on an H100, and a $2.66\times$ end-to-end Qwen2.5-7B prefill speedup at 64K-token context.

Compiler Optimizations for GPU Databases

Research Intern, with Dr. Kaushik Rajan, Microsoft Research May 2024 – Aug 2024

- Extended an MLIR-based compiler framework for GPU databases.
- Characterized performance bottlenecks and identified optimization opportunities.

UniSparse: An Intermediate Language for General Sparse Format Customization

Ph.D. candidate, with Prof. Zhiru Zhang, Cornell University Dec 2021 – Dec 2023

- Proposed a unified tensor-format abstraction that expresses both classic and custom hardware-friendly sparse formats.
- Implemented an MLIR-based compiler flow that automatically converts tensor formats and generates sparse processing kernels.

A Programming Abstraction for Reconfigurable Dataflow Architectures

Ph.D. Intern, with Dr. Gokcen Kestor, PNNL May 2023 – Oct 2023

- Designed a programming abstraction for reconfigurable dataflow architectures.
- Implemented an MLIR-based dataflow dialect that lowers front-end linear-algebra programs to the SambaNova programming interface.

Automatic Loop Scheduling for Spatial Architectures

Compiler Intern, with Dr. Stephen Neuendorffer, AMD Xilinx, Inc. May 2021 – Aug 2021

- Formulated an ILP model that solves for optimal mappings from algorithmic loops to spatial architectures under given resource constraints.
- Implemented an automatic loop-scheduling flow driven by the ILP solver as part of the MLIR AEngine toolchain.

Backend Optimization for HeteroCL

Ph.D. student, with Prof. Zhiru Zhang, Cornell University Sep 2020 – May 2021

- Implemented a loop-unrolling pass at the LLVM IR level for HeteroCL.
- Built a Mentor Graphics Catapult High-Level Synthesis (HLS) backend for HeteroCL.

MatRaptor: A Sparse-Sparse Matrix Multiplication Accelerator Based on Row-Wise Product

Ph.D. student, with Prof. Zhiru Zhang, Cornell University Nov 2019 – Sep 2020

- Co-designed a hardware-friendly sparse format and a row-wise-product-based accelerator architecture for sparse-sparse matrix multiplication (SpGEMM).
- Implemented the accelerator using PyMTL and the Mentor Graphics Catapult HLS tool.

Systolic Array Architectures for Matrix Decomposition Algorithms

Undergraduate Research Intern, with Prof. Jason Cong, UCLA Jul 2018 – Oct 2018

- Designed highly optimized 1D and 2D systolic-array architectures based on polyhedral analysis using Xilinx Vivado HLS.
- Achieved up to $50.13\times$ and $4.58\times$ higher throughput than the Xilinx HLS linear-algebra library and LAPACK, respectively, on single-threaded CPUs.

PROFESSIONAL SERVICES

Journal Reviewer

- The Journal of Supercomputing | Springer Nature 2026
- ACM Transactions on Architecture and Code Optimization (TACO) 2023, 2025, 2026
- IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD) 2023
- ACM Transactions on Reconfigurable Technology and Systems (TRETs) 2023, 2024

▪ IEEE Journal of Biomedical and Health Informatics (JBHI) 2024

TEACHING EXPERIENCE

Teaching Assistant

▪ ECE 2400: Computer Systems Language Spring 2024

HONORS & AWARDS

Jacobs Scholar Fellowship, Cornell University 2019
Excellent SRT (Student Research Training) Project Award, Tsinghua University 2019
Summer Overseas Research Training Scholarship, Tsinghua University 2018

PUBLICATIONS

CONFERENCES

- [1] J. Liu, Z. Zhao, Z. Ding, B. Brock, H. Rong, and Z. Zhang, **UniSparse: An Intermediate Language for General Sparse Format Customization**, *Proceedings of the ACM on Programming Languages*, Vol. 8, OOPSLA1, Article 99 (April 2024), 29 pages. *Object-Oriented Programming, Systems, Languages, and Applications 2024 (OOPSLA'24)*, October 20-25, 2024, Pasadena, California, USA. DOI: 10.1145/3649816.
- [2] N. Srivastava, H. Jin, J. Liu, D. Albonesi and Z. Zhang, **MatRaptor: A Sparse-Sparse Matrix Multiplication Accelerator Based on Row-Wise Product**, *2020 53rd Annual IEEE/ACM International Symposium on Microarchitecture (MICRO'20)*, Athens, Greece, 2020, pp. 766–780, DOI: 10.1109/MICRO50266.2020.00068.
- [3] J. Liu, H. Pu, and Z. Zhang, **AsyncSparse: Accelerating Sparse Matrix-Matrix Multiplication on Asynchronous GPU Architectures**, *arXiv preprint arXiv:2604.17834 [cs.DC]*, 2026.

JOURNALS

- [1] J. Liu, Z. Zhao, Z. Ding, B. Brock, H. Rong and Z. Zhang, **An Intermediate Language for General Sparse Format Customization**, in *IEEE Computer Architecture Letters*, vol. 22, no. 2, pp. 153–156, July-Dec. 2023, DOI: 10.1109/LCA.2023.3262610.
- [2] H. Chen, Z. Cao, S. Su, J. Liu and Z. Wang, **Measurement System for Attitude of Anterior Pelvic Plane and Implantation of Prosthesis in THR Surgery**, in *IEEE Transactions on Instrumentation and Measurement*, vol. 67, no. 8, pp. 1913–1921, Aug. 2018, DOI: 10.1109/TIM.2018.2809818.
- [3] H. Chen, Z. Yang, J. Zhang, J. Liu, H. Tang, Y. Zhou, B. Zhu, Z. Wang, **An IMU-Based Real-Time Measuring System for Acetabular Prosthesis Implant Angles in THR Surgeries**, in *IEEE Sensors Journal*, vol. 21, no. 17, pp. 19407–19415, 1 Sept.1, 2021, DOI: 10.1109/JSEN.2021.3091583.

POSTERS

- [1] Jie Liu, Jason Cong, **Dataflow Systolic Array Implementations of Matrix Decomposition using High Level Synthesis**, *27th ACM/SIGDA International Symposium on Field-Programmable Gate Arrays (FPGA 2019)*, February 24-26, 2019, Monterey Bay Seaside, California, USA.
- [2] Jie Liu, Hong Chen, Zhihua Wang, **IMU-based Real-Time Acetabular Prosthesis Implant Angles Measurement in Total Hip Replacement Surgeries**, *IEEE Biomedical Circuits and Systems Conference (BioCAS 2018)*, October 17-19, 2018, Cleveland, Ohio, USA.

PRESENTATIONS

- [1] Jie Liu, Zhongyuan Zhao, Zijian Ding, Benjamin Brock, Hongbo Rong, and Zhiru Zhang, **UniSparse: An Intermediate Language for General Sparse Format Customization**
 - *The ACM SIGPLAN International Conference on Systems, Programming, Languages and Applications: Software for Humanity (SPLASH'24)* Oct 2024
 - *The Semiconductor Research Corporation (SRC) TECHCON* Sep 2023
 - *The Workshop on Distributions, Relational Algebra, Graphs, Semi-Rings, Tensors, and All That (DRAGSTERS) of the 44th ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI'23)* Jun 2023
- [2] Jie Liu, Jason Cong, **Dataflow Systolic Array Implementations of Matrix Decomposition using High Level Synthesis**
 - *The 27th ACM/SIGDA International Symposium on Field-Programmable Gate Arrays (FPGA'19)* Feb 2019